Structuring data

Dr Patricia Ryser-Welch, Jan 2025

"Data is a precious thing and will last longer than the systems themselves"

Tim Berners-Lee

Websites

Blog

WikiData entries

AI and advanced statistical
datasets for model fitting

Analytical output

Multimedia content

Records of participants

Sporting events

Olympic Games Data

Medalists records

Geolocation

Journal articles

How can we structure olympic games data to meet the users' need?

Operational data

How can I capture and store data related to the games?

Analytical data

How can I model operational data for analytical purposes?

Please note we are not focusing on the management itself of the games.

https://www.adwire.lu/en/blog/how-ai-and-data-are-leveraged-for-the-paris-olympics

How can we structure olympic games data to meet the users' need?

Flat tables

RDFs

JSON

WikiData

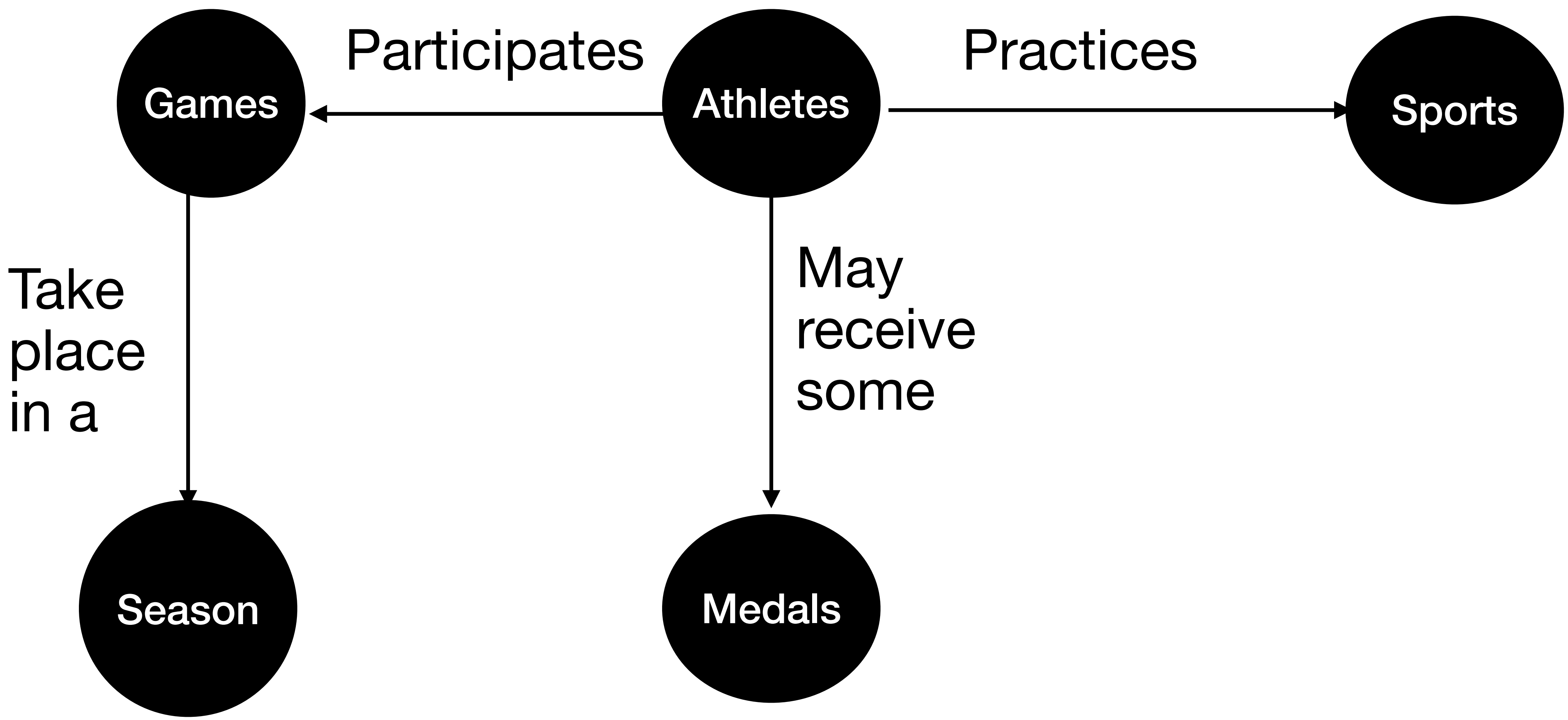
Relational databases

OLAP

Metadata - data describing data

Data

Conceptual model using knowledge graphs



A flat table, stored as a css

Name	Sex	Age	Team	NOC	Games	Year	Season	City	Sport	Medal
Aaron Michael Miller	M	30.0	United States	USA	2002 Winter	2002	Winter	Salt Lake City	Ice Hockey	Silver
Adam David Vernon Foote	M	30.0	Canada	CAN	2002 Winter	2002	Winter	Salt Lake City	Ice Hockey	Gold
Adam Enright	M	26.0	Canada	CAN	2010 Winter	2010	Winter	Vancouver	Curling	Gold
Adam Henryk Maysz	M	24.0	Poland	POL	2002 Winter	2002	Winter	Salt Lake City	Ski Jumping	Bronze
Adam Henryk Maysz	M	24.0	Poland	POL	2002 Winter	2002	Winter	Salt Lake City	Ski Jumping	Silver
Adam Henryk Maysz	M	32.0	Poland	POL	2010 Winter	2010	Winter	Vancouver	Ski Jumping	Silver
Adam Henryk Maysz	M	32.0	Poland	POL	2010 Winter	2010	Winter	Vancouver	Ski Jumping	Silver
Adam Richard Deadmarsh	M	26.0	United States	USA	2002 Winter	2002	Winter	Salt Lake City	Ice Hockey	Silver
Adelina Dmitriyevna Sotnikova	F	17.0	Russia	RUS	2014 Winter	2014	Winter	Sochi	Figure Skating	Gold
Adolf Koxeder	M	29.0	Austria-1	AUT	1964 Winter	1964	Winter	Innsbruck	Bobsleigh	Silver
Adrian Mark Aucoin	M	20.0	Canada	CAN	1994 Winter	1994	Winter	Lillehammer	Ice Hockey	Silver
Adriana Johanna "Ria" Visser	F	18.0	Netherlands	NED	1980 Winter	1980	Winter	Lake Placid	Speed Skating	Silver
Adriano Frassinelli	M	28.0	Italy-1	ITA	1972 Winter	1972	Winter	Sapporo	Bobsleigh	Silver
Adrie "Ard" Schenk	M	27.0	Netherlands	NED	1972 Winter	1972	Winter	Sapporo	Speed Skating	Gold
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Adrie "Ard" Schenk	M	23.0	Netherlands	NED	1968 Winter	1968	Winter	Grenoble	Speed Skating	Silver
Adrien Louis Albert Vandelle	M	21.0	France	FRA	1924 Winter	1924	Winter	Chamonix	Military Ski Patrol	Bronze
Adrien Plavsic	M	22.0	Canada	CAN	1992 Winter	1992	Winter	Albertville	Ice Hockey	Silver
Aino-Kaisa Saarinen	F	27.0	Finland	FIN	2006 Winter	2006	Winter	Torino	Cross Country Skiing	Bronze
Aino-Kaisa Saarinen	F	31.0	Finland	FIN	2010 Winter	2010	Winter	Vancouver	Cross Country Skiing	Bronze
Aino-Kaisa Saarinen	F	31.0	Finland	FIN	2010 Winter	2010	Winter	Vancouver	Cross Country Skiing	Bronze
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Which is issue do we have with our flat table?

Name	Sex	Age	Team	NOC	Games	Year	Season	City	Sport	Medal
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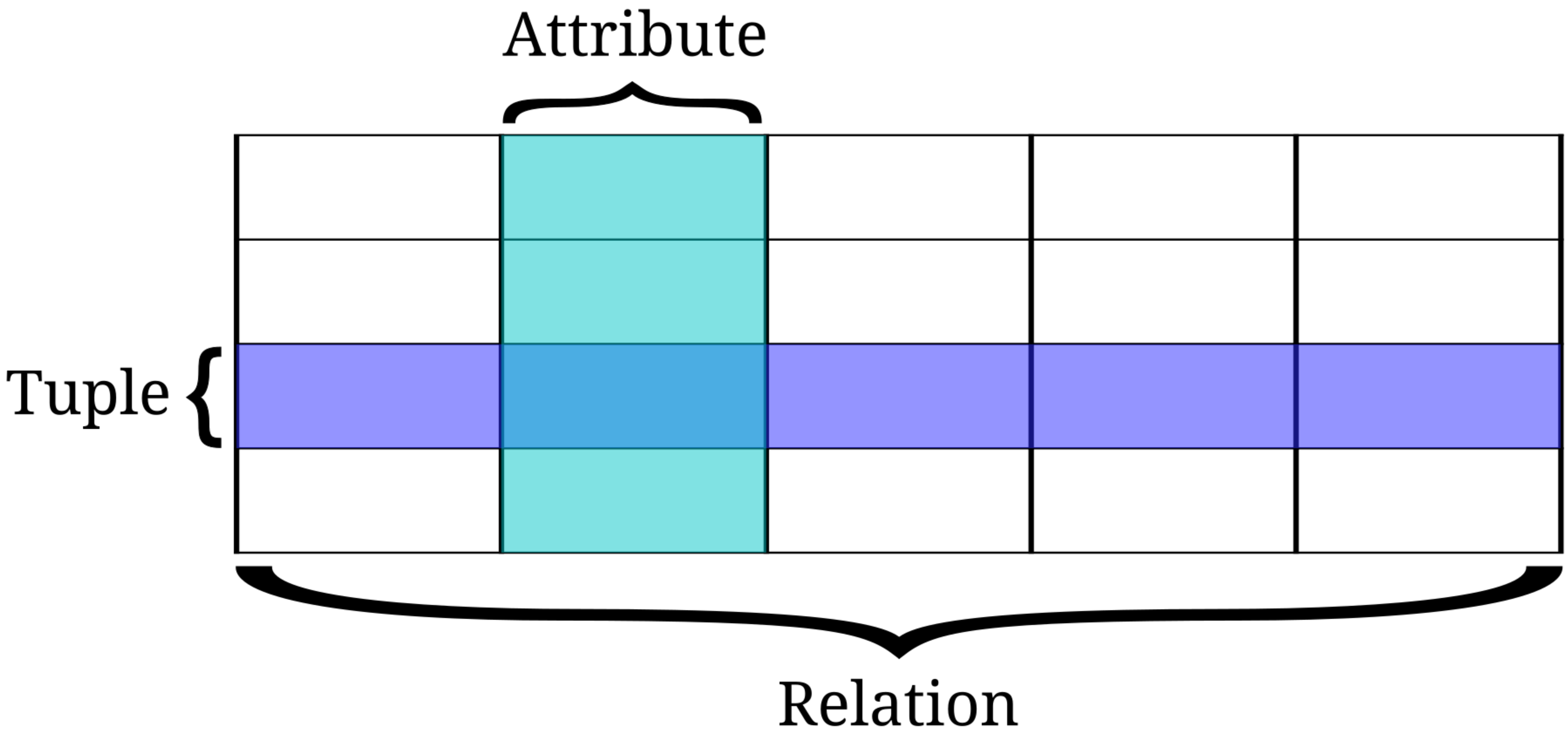
Which is issue do we have with our flat table?

- May have duplicates
- Attributes may have inaccurate domain (data types)
- Not all the rows have the same columns
- Errors may be introduced in more than one rows.
- May have data inconsistencies
- Complex data management

Removing duplication of data



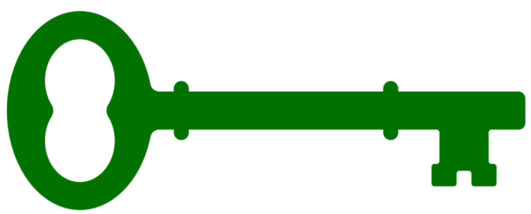
Data modelling of relational database

Concepts

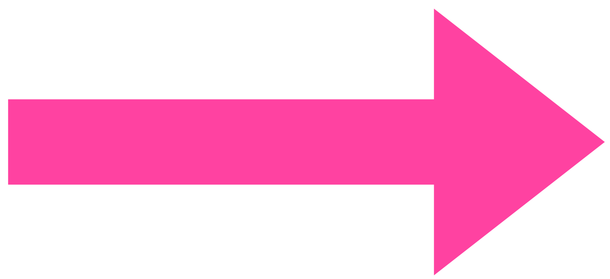
Super key

Primary key

Candidate key



Functional dependencies



Data modelling of relational database

Database Normalisation - Label from 1NF to 6NF

- Adhere to all the 5NF requirements
- a candidate key plus no more than one other (key or non-key) attribute

- It should adhere to all the 2NF requirements.
- It should only be dependent on the primary key (no transitive functional dependencies).

- Adhere to all the 1NF requirements.
- Have one primary key applied.

- Adhere to all the 4NF requirements
- the table is broken down into as many parts as possible to avoid data redundancy.

- Every cell should only have one single value.
- Every record should be unique.

- It should be in 3NF.
- X should be a super key for every functional dependency ($X \rightarrow Y$).

Data modelling of relational database

Database Normalisation

6-NF

- Adhere to all the 5NF requirements
- a candidate key plus no more than one other (key or non-key) attribute

3-NF

- It should adhere to all the 2NF requirements.
- It should only be dependent on the primary key (no transitive functional dependencies).

2-NF

- Adhere to all the 1NF requirements.
- Have one primary key applied.

5-NF

- Adhere to all the 4NF requirements
- the table is broken down into as many parts as possible to avoid data redundancy.

1-NF

- Every cell should only have one single value.
- Every record should be unique.

4-NF

- It should be in 3NF.
- X should be a super key for every functional dependency (X → Y).

Data modelling of relational database

Database Normalisation

6-NF

- Adhere to all the 5NF requirements
- a candidate key plus no more than one other (key or non-key) attribute

3-NF

- **It should adhere to all the 2NF requirements.**
- **It should only be dependent on the primary key (no transitive functional dependencies).**

2-NF

- **Adhere to all the 1NF requirements.**
- **Have one primary key applied.**

5-NF

- Adhere to all the 4NF requirements
- the table is broken down into as many parts as possible to avoid data redundancy.

1-NF

- **Every cell should only have one single value.**
- **Every record should be unique.**

4-NF

- It should be in 3NF.
- X should be a super key for every functional dependency (X → Y).

Assessment:

- **1-NF met** - every record is unique and every cell has a unique value
- **2-NF** not met - We have yet any primary keys
- **3-NF** not met - We have some transitive functional dependencies

Adam Henryk Maysz	M	24.0	Poland	POL	2002 Winter	2002	Winter	Salt Lake City	Ski Jumping	Bronze
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Using Pandas to help with data normalisation

Let’s explore further

<u>**Some possible groupings**</u>

game(year, season, city)

sports(sport)

event(event)

team(team)

athlete(name, age, gender)

noc(noc, region, notes)

medal(medal)

<u>**With Primary keys**</u>

game(id(PK), year, season, city)

sports(id (PK), sport)

event(id (PK), event)

team(id (PK), team)

athlete(id (PK), name, age, gender)

noc(noc (PK), region, notes)

medal(id(PK),medal)

Let's explore further

With relationships

game(id(PK), year, season, city)

game_athlete(game_id(PK,FK), athlete_id(PK,FK))

achievement(game_id(FK, PK), athlete_id(FK, PK), sport_id (FK,PK), event_id(FK,PK), medal_id(FK,PK), medal_ref(PK))

sports(id (PK), sport)

event(id (PK), event, sport_id (FK))

team(id (PK), team)

game_event_noc(game_id(PK,FK), event(PK, FK), NOC(PK, FK))

athlete(id (PK), name, age, gender, team_id(FK))

noc(noc (PK), region, notes)

medal(id(PK),medal)

Data Cube: Online Analytical Processing

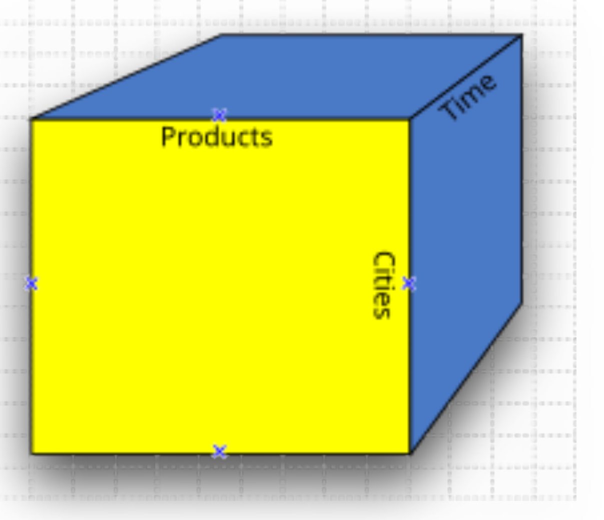
https://en.wikipedia.org/wiki/OLAP_cube

Multi-dimensional model:
- Game
- Country
- Sex
- Sport
- Medal

Numerical values:
- Age
- Number of medals

OLAP analytical model:
Dimension
- City(City (PK), latitude, longitude, country)
- Game (Name (PK), Year, Season, City(FK))
- Country (NOC (PK), Country, Team)
- Sex(Gender (PK), Gender at birth)
- Sport (Sport (PK), sport)
- Medal(Medal(PK), type)

Analytical cube
- Achievements(game(FK, PK), country(FK, PK), Sex(PK, FK), sport (PK, FK), Medal(FK, PK), Age, Number of Medals)



JSON and document-based databases

```
{
 "athlete":
      {
          "name": "Adam Henryk Maysz",
          "sport": "ski jumping",
          "sex":"M",
          “country":
          {
                 "noc": "POL",
                  "Team":"Poland"
           },
            "medals":[
                          {
                              "age":24.0,
                               "game": "2002 Winter",
                                "city": "Salt Lake City",
                                "Medal": "Bronze"
                          },
                          {
                              "age":24.0,
                              "game": "2002 Winter",
                               "city": "Salt Lake City",
                               "Medal": "Silver"
                      },
                      {
                              "age": 32.0,
                               "game": "2010 Winter",
                               "city": "Vancouver",
                               "Medal": "Silver"
                      }]
      }
}
```

JSON Schema

Copy

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "Generated schema for Root",
  "type": "object",
  "properties": {
    "athlete": {
      "type": "object",
      "properties": {
        "name": {
          "type": "string"
        },
        "sport": {
          "type": "string"
        },
        "sex": {
          "type": "string"
        },
        "country": {
          "type": "object",
          "properties": {
            "noc": {
              "type": "string"
            },
            "Team": {
              "type": "string"
            }
          },
          "required": [
            "noc",
            "Team"
          ]
        },
        "medals": {
```

https://transform.tools/json-to-json-schema

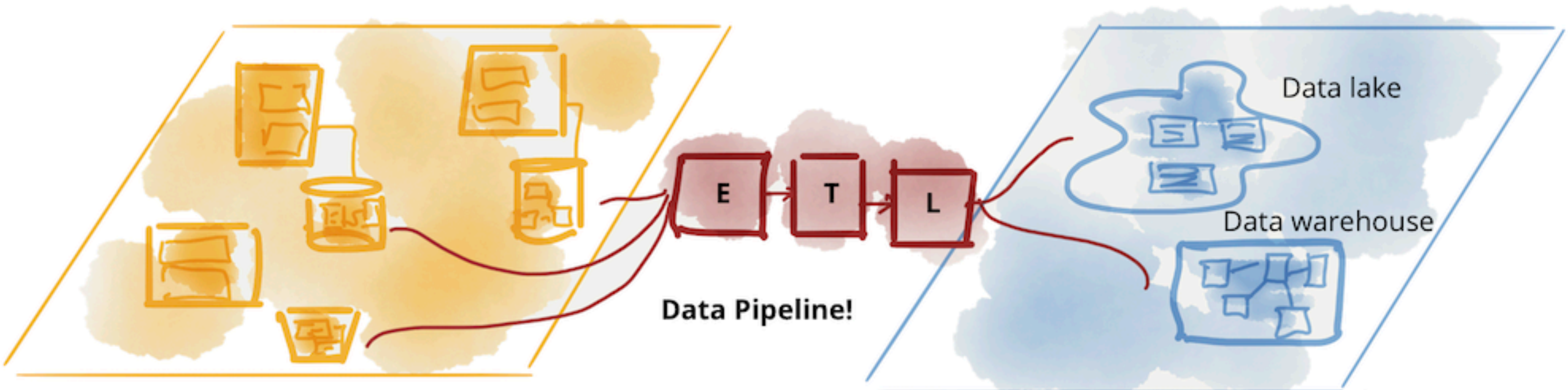
RDF

```
<?xml version="1.0"?>

<RDF>
  <Description about="https://en.wikipedia.org/wiki/
2002_Winter_Olympics">
    <year>2002</year>
    <city> https://www.google.com/maps/place/Salt+Lake+City,+UT,
+USA<city>
   <Achievements>
      <sport> https://en.wikipedia.org/wiki/Ski_jumping</sport>
      <medals>
         <bronze> https://en.wikipedia.org/wiki/Adam_Małysz</bronze>
         <silver> https://en.wikipedia.org/wiki/Adam_Małysz</silver>
      </medals>
   </Achievements>
  </Description>

</RDF>
```

Operational data vs Analytical data



Operational Data Plane

Running the business
Serving the users

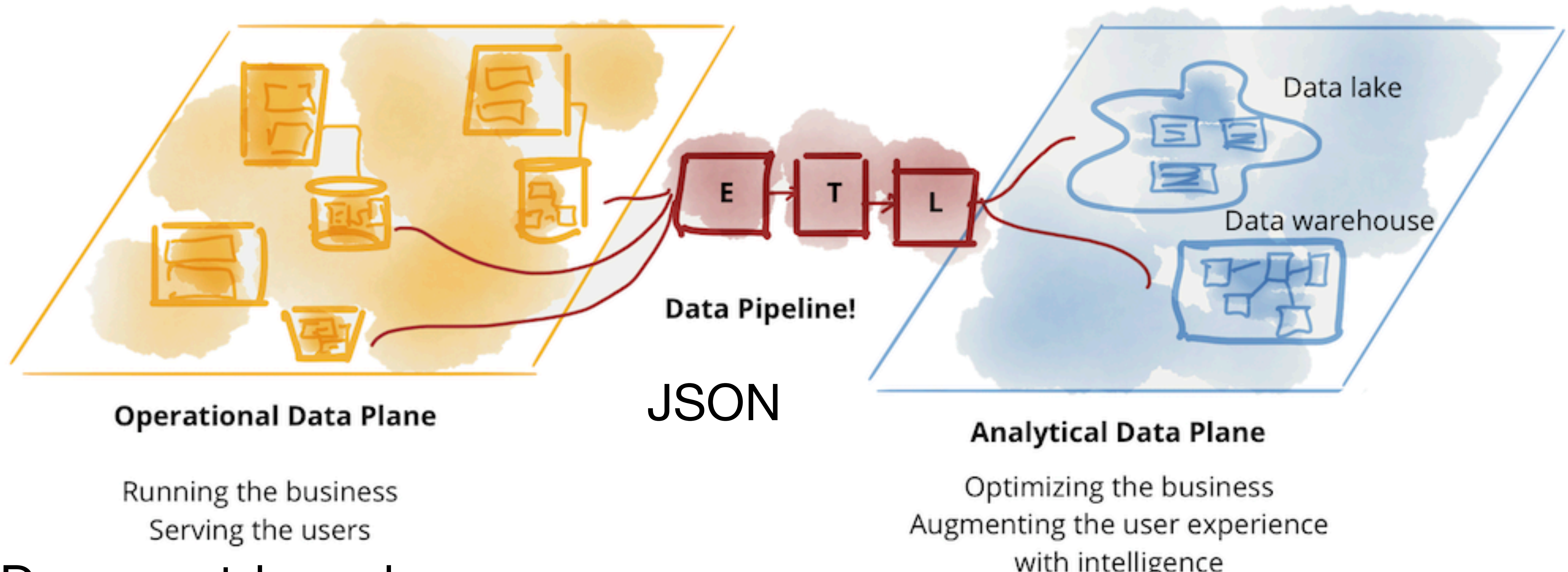
Analytical Data Plane

Optimizing the business
Augmenting the user experience
with intelligence

Operational data vs Analytical data

Relational database

OLAP



JSON

Document-based
Database

RDF

RDF

Document-based
Database

Some use of analytical data

https://www.kaggle.com/code/patriciaryserwelch/predicting-medals

https://www.kaggle.com/code/sashacooper/olympics-historical-analysis-sql-analysis

The end ….